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#2: The Kalman Filter (/controlblog/the-kalman-filter)

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This is an excerpt from an eBook on Sensor Fusion and Tracking that I wrote for Mathworks (https://www.mathworks.com). *They have graciously let me reproduce a portion of it for this blog. It has been modified and updated to improve clarity and to allow it to stand on its own apart from the rest of the eBook. You can download the full eBook here:*

https://www.mathworks.com/campaigns/offers/understand-multi-object-tracking.html (https://www.mathworks.com/campaigns/offers/understand-multi-object-tracking.html)

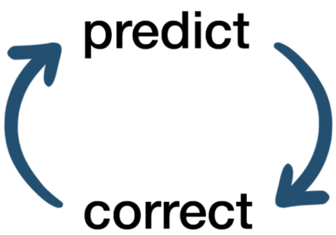
This is a long post. Here is the tl;dr for those in a hurry!

A Kalman filter is an algorithm that we use to estimate the state of a system. It does this by combining a noisy measurement from a sensor with a flawed prediction from a process model. If the measurement noise can be modeled as a Gaussian distribution and the process model can be modeled as linear with a Gaussian error distribution, then a linear Kalman filter will produce an optimal state estimate. The optimal estimation turns out to be the product of the two Gaussian distributions. Therefore, the linear Kalman filter equations can be thought of as a prediction step plus Gaussian multiplication. And I think that's pretty awesome!

Ok, now on to the full post.

There are many reasons why we want an accurate estimate of the state of a system. One of which is that we need it in feedback control to determine how to change the system. For example, if we are trying to control the temperature of something, we need to have a good estimate of what the current temperature is in order to know if we need to add heat or remove heat from the system.

It's probably pretty obvious that directly measuring the system with one or more sensors is a way to estimate state, however, what might not be obvious is that we can usually improve our estimate by supplying additional information beyond just what the sensor itself is measuring. One of the methods we have for using some of that additional information is with a Kalman filter.

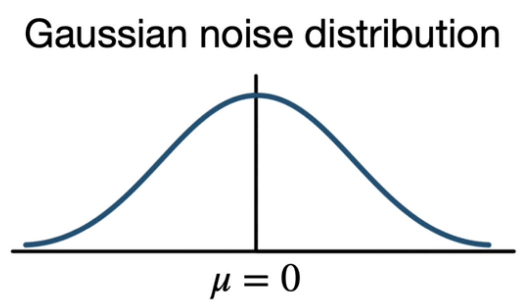


A Kalman filter is part of a class of estimation filters that use a two-step process, i.e prediction and correction, to produce an optimal state estimate.

If the system can be described with a **linear model** and the probability distribution of the process noise and measurement noise is **Gaussian** then a *linear Kalman filter* will produce the optimal state estimation.

Linear system model

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$



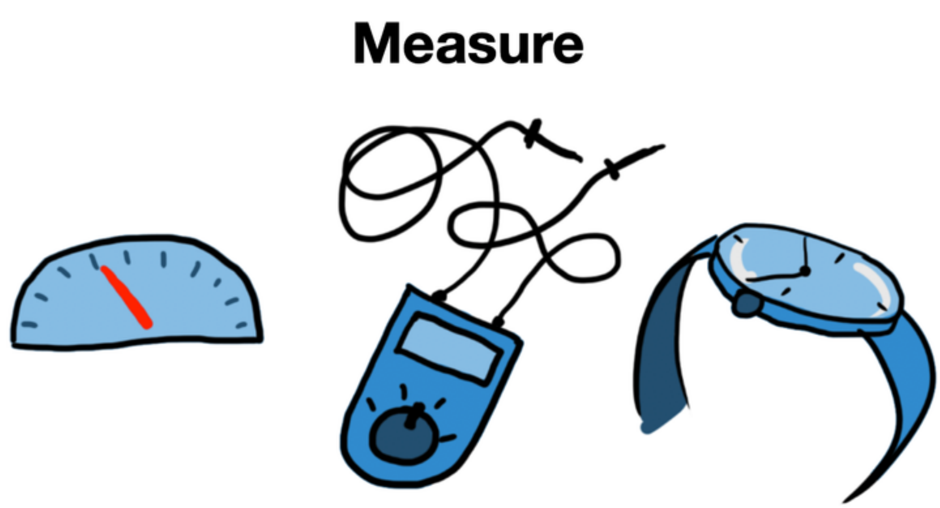
In practice, though, no real system is truly linear and noise might not have a perfect Gaussian probability distribution which means the result of a linear Kalman filter is never absolutely optimal for real systems. However, they still work very well for many real-life problems and the intuition you gain from understanding linear Kalman filters can help you to better understand their nonlinear counterparts like the extended Kalman filter, the sigma-point filter, and the particle filter.

Don't worry if everything I just described doesn't mean much to you right now because I'll touch on some of it a bit more throughout this post. With that being said, let's walk through a minimal math approach to understanding the linear Kalman filter.

HOW DO PEOPLE ESTIMATE STATE?

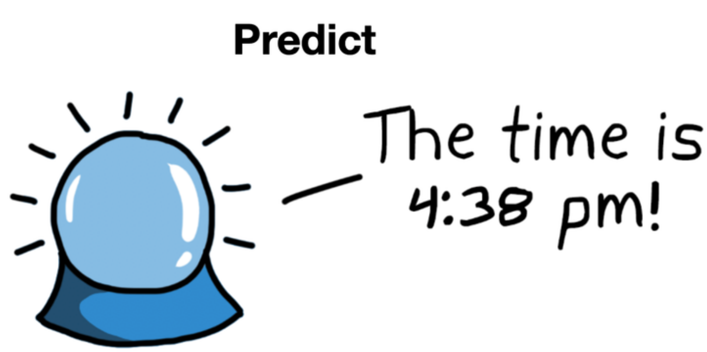
We can start by thinking about how we as humans estimate the state of something. For example, how do we know the velocity of a vehicle, the voltage of an electrical circuit, or the time of day?

The obvious answer is that we measure it directly with a sensor like we do with a speedometer, a voltmeter, or a clock. Similarly, we may derive the state we want by measuring other quantities. This would be the case if we estimate *velocity* using the difference between two successive *position* measurements.



But, as humans, we have other knowledge that we use to challenge the validity of our measurements. We have mental models of the world and we use those models to create an expectation of what we're measuring. And if our expectation differs from the measurement we go through a process of trying to determine which result we trust more. If we trust both results somewhat, but neither fully, then we might blend them together to produce a best guess.

For example, if you look at your watch and it says it's 4:37 pm and then you wait about a minute, check again, and it says 6:15 pm, you wouldn't automatically trust that result. We have this general understanding of the passage of time and how the universe should behave and this extra knowledge allows us to make a prediction of how the state of a system should change over some time period. In this case, we would assume our sensor (the watch) was faulty and we'd put more trust in our estimate of the time.



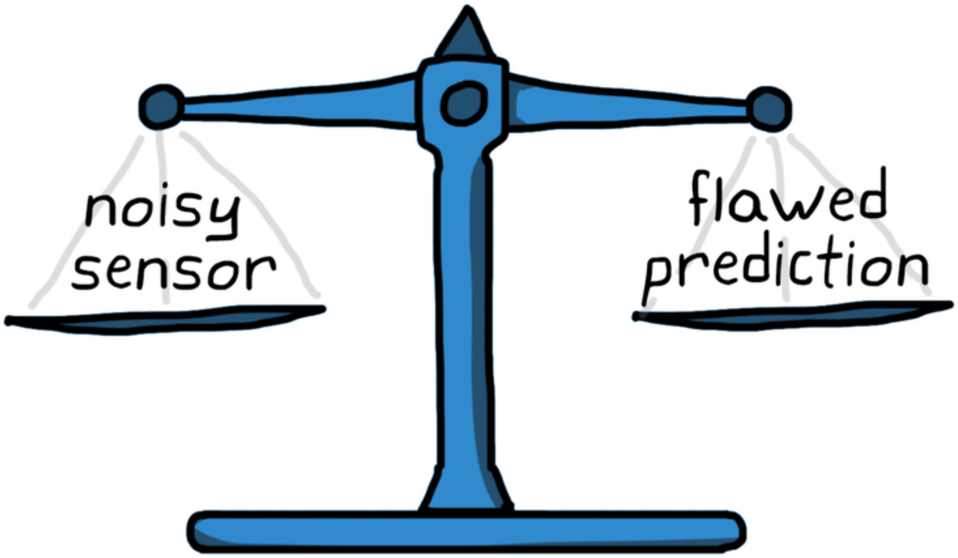
IF OUR MODEL IS GOOD, WHY USE MEASUREMENTS AT ALL?

If we have a perfect model and we knew the initial condition of our system, then we would be able to predict how the system state will change over time, perfectly. I mean, why mess around with sensors if we know everything? Well, unfortunately, our ability to predict the outcome of something isn't perfect, it's subject to error. This error comes from us having the wrong model of the system and from not knowing and accounting for all of the external inputs that can influence the system state. Therefore, it's not wise to base the state of the system solely on our flawed prediction. Typically, the longer we predict into the future, the less certain we are in the answer since errors tend to build up over time.

Continuing with the watch example, if you checked your watch and then waited for what you thought was an hour but according to your watch had only been 54 minutes, then you are probably more inclined to believe the measurement and discount your prediction.

This is exactly the thought process behind a Kalman Filter. It combines (in an optimal way) the measurements from a noisy sensor *and* the prediction from a flawed model to get a more accurate estimate of system state than either one independently could provide.

So, how does the Kalman filter accomplish this?



LET'S SEE SOME MATH!

These are the discrete-time equations for a linear Kalman filter. At first glance, they seem complicated and if you're not familiar with the mathematical notation, then maybe also a bit nonsensical.

They are separated into predict (using the model) and correct (using the measurements), hence the two-step process for estimating state. The predict step is responsible for propagating the state vector into the future using the linear model, and the correct step blends the current prediction with a current measurement to get the corrected estimated state.